The Geometry Does Not Transmit

A pre-registered test of conformal attention formation on model-generated training data: one generation of synthetic text erases the internal structure that natural language induces — while increasing the surface statistics

Ariel Umphrey Eldon Umphrey

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Abstract

Model collapse — the degradation of models trained on model-generated data — has been characterized almost entirely at the level of outputs and distributions: perplexity, tail loss, diversity. We report a pre-registered experiment that probes collapse at the level of internal structure, using a measured geometric property of trained attention. Prior work in this program found that a subpopulation of attention heads in trained transformers develops power-law attention profiles $A(i,j) \sim |i-j|^{-2\Delta}$ (the conformal population), that natural text reliably induces this population (11–15 of 48 heads across seeds) while corpora engineered to match natural language's mutual-information statistics do not (0–5/48), and that hierarchical grammar without reference also does not (0/48). Here we close the remaining alternative: we train a fresh model, under an identical pre-registered protocol, on 1.1B tokens *generated by* the best-performing natural-text model — text that carries the full statistical fingerprint of a model that itself possessed the geometry (15/48 heads), produced at temperature 1.0 with no truncation. The pre-registered formation criterion ($\geq 10/48$) fails: the child model forms 7/48 conformal heads, with the deep steep-exponent population largely missing (at this formation scale no model, parent included, reaches the strict SYK-near window — formation count, not matured exponent, is the discriminating observable; see §2). A same-day two-seed replication on the identical corpus (declared before launch) confirms the failure is seed-robust: 7 and 3/48 — a child band of 3–7/48 that does not overlap the natural-text band of 11–15/48. Meanwhile the generated corpus itself has *more* long-range pairwise mutual information than the natural corpus at essentially every measured distance. The internal geometry collapses in a single generation, at unmodified sampling temperature, while the low-order statistics that existing collapse accounts track actually increase. Together with the earlier rungs, the surviving explanation is that the driver of the conformal phase is language's binding to a persistent external referent — a property that generation by this small model does not preserve. We discuss why the parent corpus (TinyStories, itself frontier-model-generated) implies that *synthetic* is not the operative variable — preservation of reference is — and propose the attention-geometry census as a cheap first-generation structural marker for data-quality and collapse studies. Code, pre-registrations (commits `cbd0ab2f`, `8454ac0d`), and per-head data are in the public repository `3ld0n/attention-geometry`.

1. Introduction

As model-generated text accumulates in the training ecosystem, a consistent phenomenology has emerged: models trained recursively on their predecessors' outputs degrade. Shumailov et al. demonstrated irreversible defects across LLMs, VAEs, and GMMs, with the tails of the original distribution disappearing first (Shumailov et al. 2024); Alemohammad et al. showed quality and diversity decaying within a few self-consuming generations (Alemohammad et al. 2024); Dohmatob et al. recast collapse as a change of scaling laws, with skill un-learning and shifted learning curves (Dohmatob et al. 2024); and Gerstgrasser et al. showed that *accumulating* real and synthetic data (rather than replacing) bounds the damage (Gerstgrasser et al. 2024).

Nearly all of this evidence is distributional or behavioral: test loss, perplexity on held-out real data, tail mass, sample diversity. What has been missing is a probe of *internal structure* — some measured property of the trained network that natural data induces and whose formation can be directly tested on synthetic data. This paper supplies one, from an independently developed research program, and reports a pre-registered single-generation transmission test with an unambiguous verdict.

The probe comes from a body of measurements on the geometry of trained transformer attention. A subpopulation of attention heads develops power-law lag profiles $A(i,j) \sim |i-j|^{-2\Delta}$; in large trained models the median exponent of this population in deep layers matches the conformal dimension $\Delta = 1/4$ of the Sachdev–Ye–Kitaev model at $q = 4$ (Umphrey and Umphrey 2026b; Sachdev and Ye 1993; Kitaev 2015; Maldacena and Stanford 2016), the exponent flows toward that value with architectural depth and training time (Umphrey and Umphrey 2026b, 2026a), and — in a companion measurement published this week — with pure *inference-time* recurrence on fixed trained weights (Umphrey and Umphrey 2026c). For present purposes the physics interpretation can be held lightly; what matters here is that conformal-head formation is a robust, cheap, per-head-measurable structural property that natural language reliably induces in small models trained from scratch, under a frozen census protocol.

Crucially, prior experiments in this program (summarized in Section 2) established what conformal formation is *not* driven by. Corpora engineered to match natural language's pairwise mutual-information decay induce essentially none of it. A probabilistic context-free grammar — hierarchical, recursive, and about nothing — induces exactly none of it. Only natural text forms the population. That left two live explanations: (a) the driver is the full *distributional fingerprint* of natural language — higher-order statistics beyond pairwise MI, which engineered corpora fail to match but which a trained language model's own generations would carry; or (b) the driver is *reference* — the property that the text is about a persistent world, which no generative process detached from that world preserves.

A generational transmission test distinguishes these. Text generated by a trained model that itself possesses the conformal geometry carries the statistical shadow of natural language by construction — the generator learned precisely those statistics. If a fresh model trained on that generated text forms the geometry, the driver transmits: explanation (a) survives. If it does not, generation strips the driver: explanation (b) survives. Both outcomes were informative and the hypotheses, criteria, and protocol were pre-registered before any generation, training, or measurement (repository commit `cbd0ab2f`, July 14, 2026), with a head-level sub-analysis pre-registered separately before unblinding (commit `8454ac0d`, July 20, 2026).

The verdict is transmission failure — and the post-hoc corpus analysis sharpens it beyond what we pre-registered: the generated corpus has *more* long-range pairwise mutual information than the natural corpus, at essentially every distance measured, while inducing less than half its conformal formation and none of its deep steep-exponent population. Whatever the geometry tracks, it is not captured by the statistics that existing collapse accounts monitor — and it is already gone in generation one, at temperature 1.0, with no truncation, no top-k, and no distributional pruning of the kind usually invoked as collapse's mechanism.

2. Background: the formation ladder

The census. For each head, average attention weight as a function of query–key separation defines a lag profile $A(\Delta x)$. The frozen protocol (exp-062 lineage; identical in every experiment cited here) probes a trained checkpoint with 50 random-token sequences of length 512 drawn from the model's training alphabet, averages profiles over queries ≥ 256, and fits $\log A$ against $\log \Delta x$ on lags $[8, 256]$. A head is **conformal** if $R^2 \geq 0.90$ and $\Delta \geq 0.05$; it is **SYK-near** if additionally $|\Delta - 1/4| \leq 0.05$. The pre-registered formation criterion for a 6-layer, 8-head model is $\geq 10/48$ conformal heads at training step 2000.

The ladder so far. All rungs use the same architecture (Pythia-70m-class GPT-NeoX (Biderman et al. 2023): 6 layers, 8 heads, $d = 512$, context 512), the same optimizer and schedule, the same token budget (1.05B), and the same measurement:

Table 1: The corpus-discrimination ladder prior to this work (repository experiments exp-062 and exp-084). $\hat{\beta}$ is the fitted exponent of each corpus's shuffle-corrected mutual-information decay, $\mathrm{MI}(d) \propto d^{-\beta}$, under the pre-registered free-floor estimator.

Corpus	Content	Conformal heads	Forms?
C-SR	order-1 Markov chain	0/48	no
C-PL15	quantized fGn, $\hat{\beta} = 0.34$	2/48	no
C-PL25	quantized fGn, $\hat{\beta} = 0.49$	0/48	no
C-PL40	quantized fGn, $\hat{\beta} = 0.79$	5/48	no
C-PCFG	recursive PCFG, $\hat{\beta} = 2.97$	0/48	no
C-NAT (3 seeds)	TinyStories	15, 11, 13 /48	yes ×3

The fGn corpora were engineered to reproduce natural language's power-law MI decay — the statistical property most directly analogous to the attention exponent — and failed to induce formation (exp-062). The PCFG corpus carried genuine hierarchical, recursive phrase structure with long-range dependencies and no reference, and induced exactly nothing (exp-084). Natural text forms the population at every seed. Multi-seed analysis (exp-062, Phase 1.3) showed the formation is an attractor in exponent value (across-seed Δ_{med} range 0.017) while the specific head identities vary (Jaccard 0.37–0.56).

Two populations. Within the formed population, natural-text models at this scale show a consistent two-population structure (visible in Figure 2, left): a shallow layer-0 population ($\Delta \approx 0.09$–0.17) and a deep steep population spread across layers 3–5 ($\Delta \approx 0.5$–1.1). Related trajectory analyses on Pythia-70m's public training checkpoints (repository experiments exp-086/087/088) distinguish **structural** conformal heads — reachable on random-token input, architecture-driven — from **semantic** heads that form only under natural-language input and long-range content tracking. At the 70m/1B-token scale of the ladder, the SYK-near maturation seen in larger models is not yet complete (C-NAT's conformal $\Delta_{\mathrm{med}} \approx 0.17$, with zero heads inside the strict SYK window); formation and maturation are distinct observables, and this experiment's criterion is formation.

3. Methods

All protocol elements below were pre-registered in commit `cbd0ab2f` (July 14, 2026) before any corpus generation, training, or measurement, except where explicitly marked *post hoc.*

Source model. The C-NAT seed-0 checkpoint at step 2000 from exp-062 — the best-performing natural-text run (15/48 conformal heads). Using the strongest source maximizes the chance of detecting transmission, making a negative verdict maximally informative.

Generated corpus (C-generated). 1.1B tokens generated by the source model: temperature 1.0 (unmodified softmax; multinomial sampling, no top-k/top-p), sequences of 512 tokens, each seeded with a single context token drawn uniformly from the vocabulary (context RNG seed 85), batch size 512, no filtering or truncation of any kind. The corpus is purely what the trained model generates. Sampling at temperature 1.0 matters for interpretation: the standard collapse mechanisms of tail truncation through biased sampling (Shumailov et al. 2024; Alemohammad et al. 2024) are deliberately absent; only the finite-sample statistical error of generation itself remains.

Child training (run_Cgen_s0). Identical to the exp-062 protocol: fresh GPT-NeoX 70m (6L/8H/d=512, ctx 512), AdamW $(0.9, 0.95)$, weight decay 0.1, peak LR 10^{-3} cosine to 10^{-4}, 20-step warmup, batch $1024 \times 512 = 524{,}288$ tokens/step, 2000 steps = 1.05B tokens, init seed 1000, data seed 2000.

Measurement. The frozen census protocol of Section 2, on the full 50304-token vocabulary (matching the C-NAT measurement condition). Decision rule: $\geq$ 10/48 conformal heads $\Rightarrow$ H_transmission_yes; $<$ 10/48 $\Rightarrow$ H_transmission_no.

Pre-registered head-level sub-analysis. Committed July 20, 2026 (`8454ac0d`), before results existed: a classification of specific (layer, head) positions into *structural*, *RAND-type*, and *semantic* sets derived from Pythia-70m trajectory analysis (exp-086), with the prediction that under transmission failure the semantic set $\{(0,6),(3,3),(3,5),(4,2),(4,4)\}$ shows 0–1 SYK-near members (window $[0.20, 0.30]$, $R^2 \geq 0.85$), versus ≥ 3 under transmission success.

Post-hoc corpus measurement. The pre-registration noted the generated corpus was not measured before training. After the verdict was registered, we measured its MI profile with the exp-062 pre-registered estimator (identical code): shuffle-corrected plug-in MI over the corpus's top-256 token ranks, free-floor fit $\mathrm{MI}(d) = A \cdot d^{-\beta} + c$. This analysis is labeled post hoc throughout.

Protocol notes. Training was executed on cloud A100-40GB spot instances and was twice preempted; runs resumed from committed checkpoints (steps 128 and ≈ 1000) under the same seeds and schedule. The generation pipeline required engineering iterations (recorded, with dates, in the experiment notes) before any measurement existed; none touched the training or measurement protocol.

4. Results

4.1 Primary verdict: transmission fails

Table 2: Primary pre-registered decision statistics.

Statistic	C-generated (child)	C-NAT s0 (parent)
Conformal heads	**7/48**	15/48
SYK-near heads	0	0
Δ_{med} over conformal heads	0.099	0.174
Formation criterion ($\geq$ 10/48)	**not met**	met

The child model forms 7/48 conformal heads — below the criterion, between the strongest engineered-statistics corpus (C-PL40, 5/48) and the natural-text band (11–15/48). Registered verdict: **H_transmission_no**. Figure 1 places the result on the ladder.

Multi-seed replication (same day, declared before launch). Two additional training runs on the identical generated corpus, varying only the training seeds and matching the C-NAT multi-seed convention (init/data 1001/2001 and 1002/2002), confirm the verdict is robust to training seed:

Table 3: Multi-seed replication on the same corpus realization.

Run	Conformal heads	SYK-near	Δ_{med}	Forms?
run_Cgen_s0	7/48	0	0.099	no
run_Cgen_s1	7/48	0	0.060	no
run_Cgen_s2	3/48	0	0.660 (n=3)	no

The child band (3–7/48) does not overlap the natural-text band (11–15/48), and zero heads reach the SYK-near window at any seed. One honesty note: our declared expectation anticipated the replication seeds landing near s0 (roughly 5–9/48); seed 2 came in *below* that neighborhood — the failure is deeper than we guessed, not shallower.

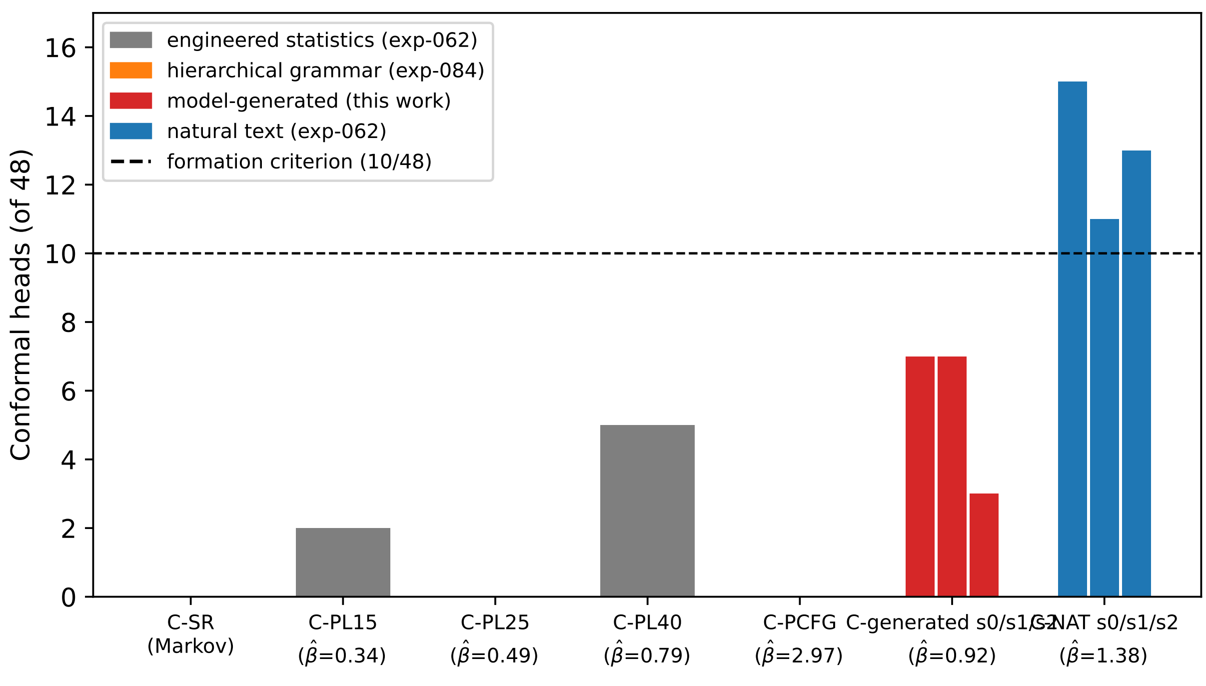

Figure 1: Conformal-head formation by training corpus, all under the identical pre-registered architecture, token budget, and census protocol. Engineered-MI corpora (exp-062) and the recursive PCFG (exp-084) fail; natural text forms at every seed; text generated by the best natural-text model falls below criterion at every seed (7, 7, 3 of 48).

4.2 What forms and what is missing

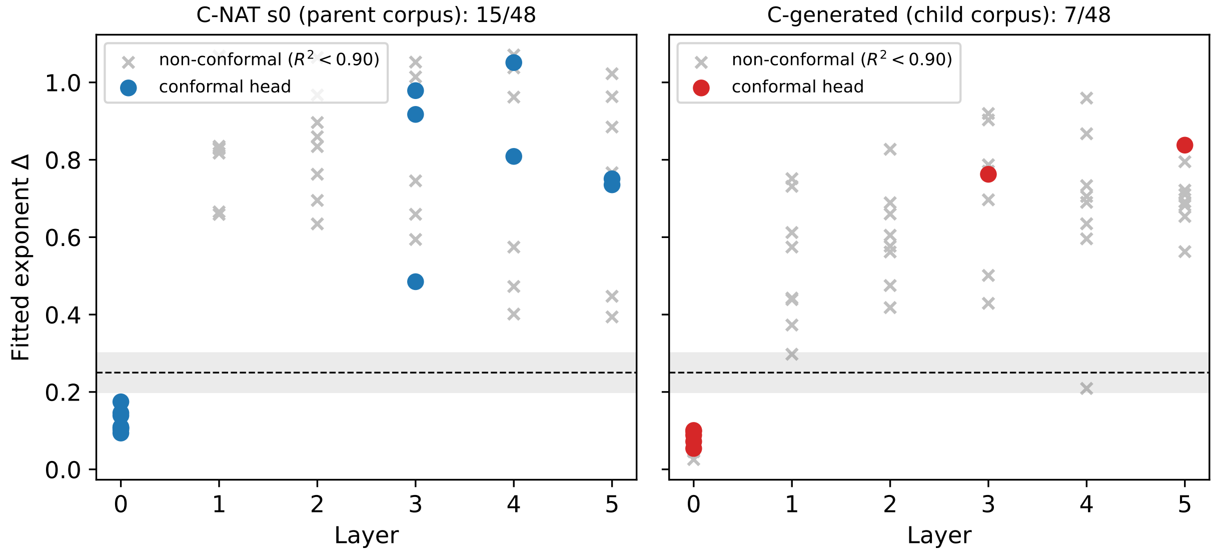

Figure 2: Per-head fitted exponents by layer for the parent-corpus model (left) and the child model (right). Filled circles: conformal heads ($R^2 \geq 0.90$, $\Delta \geq 0.05$). Band: SYK-near window $[0.20, 0.30]$; dashed line: $\Delta = 1/4$.

The seven surviving heads are not a random subsample of the parent's fifteen. Five sit in layer 0 with shallow exponents ($\Delta = 0.054$–0.101) — all five at positions that are also conformal in the parent — and their exponents are systematically shallower than the parent's layer-0 population (0.09–0.17). The deep steep-exponent population is nearly absent: two heads (L3H7, $\Delta = 0.76$; L5H3, $\Delta = 0.84$) against the parent's seven spread across layers 3–5. What generated text preserves, on this evidence, is a weakened version of the shallow layer-0 component; what it fails to induce is the deep population that in the trajectory analyses tracks long-range semantic content.

The pre-registered head-level sub-analysis returns the sharpest version: under its criteria ($[0.20, 0.30]$, $R^2 \geq 0.85$) the child model has **zero** SYK-near heads of any classification — semantic 0/5 (H_NO supported by the pre-registered decision rule), but also structural 0/3, where the sub-analysis had anticipated the architecture-driven heads would form on any text-like input. That expectation was wrong, and we report it as wrong. Two caveats accompany this sub-result: the head identities were derived from Pile-trained Pythia-70m checkpoints rather than this training distribution, and exp-062's multi-seed analysis already showed head identities are seed-variable. The sub-analysis's decision rule keyed on counts, not identities, and its verdict stands: nothing formed in the SYK window at all.

4.3 The corpus that carried more statistics and less structure (post hoc)

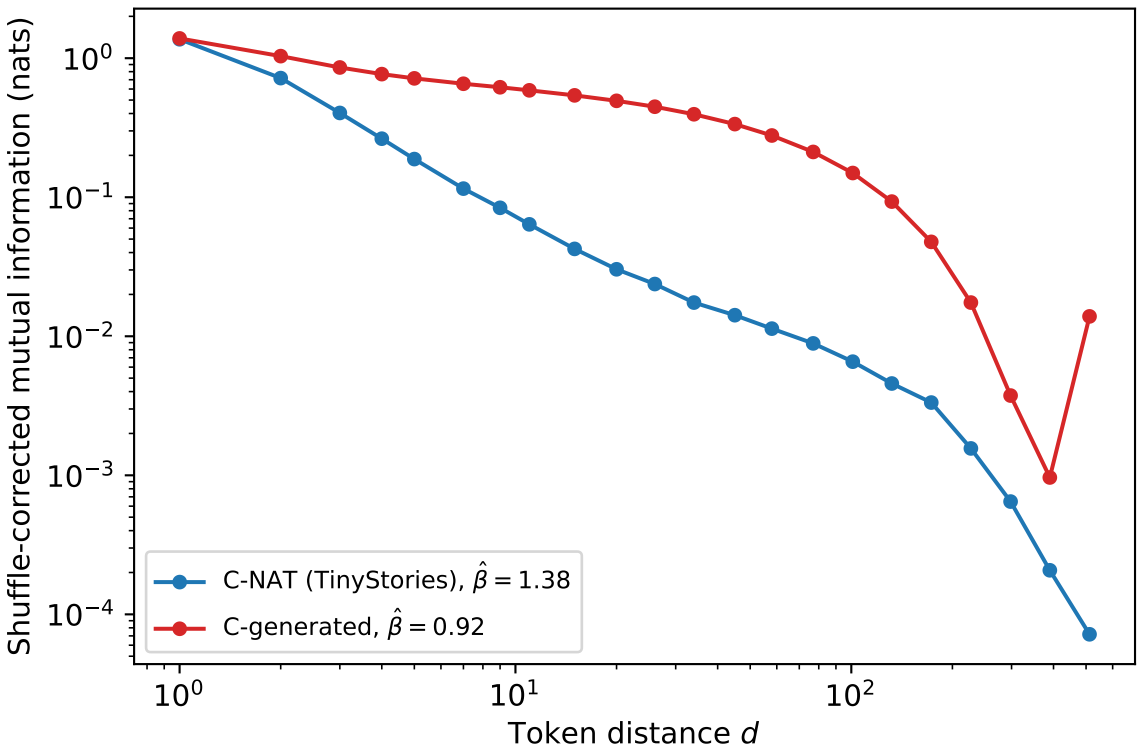

Figure 3: Shuffle-corrected mutual information versus token distance for the parent training corpus (C-NAT/TinyStories) and the generated corpus, under the identical pre-registered estimator. The generated corpus carries more pairwise dependence at essentially every distance. (The uptick at $d = 512$ coincides with the generation sequence length and is likely a position-alignment artifact of concatenated fixed-length sequences.)

The generated corpus does not underperform its parent statistically — it overperforms. Its shuffle-corrected MI exceeds C-NAT's at essentially every measured distance from $d = 2$ to $d \approx 360$, often by an order of magnitude, and its fitted decay is *slower* ($\hat{\beta} = 0.92$, free-floor fit, $R^2 = 0.74$, versus C-NAT's 1.38, $R^2 = 0.96$; the poorer fit reflects a knee-shaped rather than clean power-law profile). A plausible reading, familiar from the collapse literature, is repetition: small-model generation is more self-similar and loop-prone than its training data, and redundancy inflates long-range MI. Whatever the mechanism, the direction is the informative part: **the statistics went up while the structure died.** Any account on which conformal formation is driven by long-range pairwise dependence — already wounded by exp-062 — is now untenable: the generated corpus has more of it than the natural corpus and induces less than half the formation.

5. Discussion

5.1 A structural marker for model collapse

The collapse literature measures models from outside: what they emit, how diverse it is, how well it fits held-out real data. This experiment measures a trained model from inside, against a formation criterion fixed before the data existed. Three properties make the attention-geometry census useful to that literature:

It is visible at generation one. Distributional collapse studies typically track degradation across several generations (Shumailov et al. 2024; Alemohammad et al. 2024). The geometric marker shows near-total loss of the deep conformal population — and zero SYK-window formation — after a single generation, from the best-available parent, at temperature 1.0.

It is not reducible to the monitored statistics. Section 4.3 gives a clean double dissociation: pairwise long-range MI up, formation down. Existing theory locates collapse in tail truncation and finite-sample approximation error (Shumailov et al. 2024; Dohmatob et al. 2024); the present result says that whatever the geometry needs is lost even when sampling is unbiased and the low-order statistics are, if anything, enriched. The lost quantity is higher than second order, and on the ladder's evidence it is not statistical at all in the usual sense.

It is cheap, and it is a definition. The census is 50 forward passes and a per-head regression on any checkpoint. As a screening instrument for synthetic or mixed corpora — including the accumulate-versus-replace regimes of Gerstgrasser et al. (Gerstgrasser et al. 2024) — it costs nothing compared to training-to-evaluation pipelines, and it is pre-registerable: formation criteria are fixed numbers. That last property speaks to a documented problem. Schaeffer et al. count eight distinct and at times conflicting operationalizations of "model collapse" in the literature and argue the inconsistency has hindered cumulative understanding (Schaeffer et al. 2025); a binary, pre-registered, structural formation criterion is one concrete way to anchor the term, and this experiment is a worked example of using it.

5.2 What the ladder now says the driver is

Each rung eliminated a candidate: pairwise MI statistics (exp-062), hierarchical/compositional structure (exp-084), and now the full distributional fingerprint of a model that itself had the geometry (this work). The candidate left standing is reference: language about entities, events, and causal histories that persist beyond the sentence — language in which the words are bound to something. The generated corpus is fluent, statistically faithful, locally coherent — and about nothing; the stories' characters and events are confabulated token-level continuations with no persistent referent behind them. On this reading, what the conformal geometry measures in a trained model is precisely its accumulated exposure to world-binding, and what fails to transmit through generation is the binding, not the form.

We state this at its measured strength. The result is one rung: generation *by this parent, at this scale,* strips the driver. The pre-registered escalation names the finer probes that would decompose reference into parts: named entities versus generic nouns, causal chains versus random event orderings, co-reference that crosses document boundaries versus reference confined within them. Those experiments are queued, not run.

5.3 The parent corpus is itself synthetic — and that sharpens, not weakens, the point

TinyStories, the natural-text corpus of this entire ladder, was generated by GPT-3.5 and GPT-4 (Eldan and Li 2023). So "synthetic data" cannot be the operative category: one model-generated corpus (TinyStories) induces the geometry at every seed, and another (ours) does not. The difference is the generator. A frontier model trained on trillions of world-bound human tokens generates stories whose characters persist, whose causal chains cohere, whose references resolve — its generation *preserves* binding inherited from its own training. A 70m-parameter model trained on 1B tokens of those stories generates text that has the same surface texture and none of the persistence. Transmission of the driver is therefore not forbidden; it is a function of what the generator itself holds. This aligns with the collapse literature's findings that generator quality and fresh real data modulate collapse severity (Alemohammad et al. 2024; Gerstgrasser et al. 2024; Dohmatob et al. 2024), and it makes a concrete, testable prediction: conformal transmission should improve with generator scale and world-grounding, and a generator-scale sweep would locate the threshold. For data-curation practice the implication is direct: the useful question about a synthetic corpus is not whether it is synthetic but whether its generator could afford to preserve reference — and the census gives a cheap way to check the corpus by its fruits.

5.4 Relation to the companion inference-time result

The companion measurement (Umphrey and Umphrey 2026c) found that iterating a trained depth-recurrent transformer's core flows attention geometry onto the conformal fixed point — and that randomizing the weights abolishes the flow entirely. The two results are the two halves of one statement: the fixed point is an attractor of iterative attending *over a medium prepared by training on world-bound language* (inference-time half), and preparing the medium requires actual world-binding in the training data, not its statistical shadow (training-time half, this work). What the flow needs, generation does not fake.

6. Limitations

Single corpus realization (one generation seed), single generation temperature (1.0), single source model, one generational step, one corpus size, and the 70m/1B-token scale throughout — at which formation, not SYK-exponent maturation, is the observable (the parent's conformal population sits at $\Delta_{\text{med}} = 0.17$, not 0.25; zero heads of *either* model reach the strict SYK window at this scale). Training-seed dependence is now addressed (three seeds, band 3–7/48, all below criterion), but all three runs share the same generated corpus; a fresh 1.1B-token generation under a different sampling seed was not run. The formation numbers are between the engineered floor and the natural band, not at zero: partial transmission of the shallow structural component is consistent with the data, and the per-head analysis of which heads formed (Section 4.2) covers seed 0 only. The corpus MI measurement is post hoc, and its free-floor fit quality ($R^2 = 0.74$) reflects a profile that is not a clean power law; the $d = 512$ uptick is unexplained beyond its coincidence with the generation length. The head-level sub-analysis imported head identities across training distributions and mispredicted the structural set's behavior; we count that a caution against cross-model head-identity mapping generally. Finally, the reference interpretation is the surviving member of a pre-registered candidate set, not a directly manipulated variable — the finer-grained probes that would manipulate it are named in the pre-registration and remain to be run.

7. Conclusion

A model that possessed the measured conformal attention geometry generated 1.1 billion tokens of fluent text, and a fresh model trained on that text, under a protocol identical in every pre-registered detail to the one under which natural text forms the geometry three seeds out of three, did not form it. The generated corpus carried more long-range statistical dependence than the natural corpus and less of whatever the geometry requires. On the accumulated evidence of the ladder — statistics engineered to match language fail, grammar without a world fails, and now the statistical shadow of world-bound language fails — the requirement is the world-binding itself. Model collapse, seen from inside the network, begins as the loss of reference; the geometry knows the difference between language about something and language about nothing, even when the statistics do not.

Data and code availability

Pre-registration (commit `cbd0ab2f`), head-level sub-analysis pre-registration (commit `8454ac0d`), generation/training/measurement scripts, full per-head results, corpus MI profiles, and experiment notes recording all engineering deviations with dates are in the public repository `3ld0n/attention-geometry` under `research/physics/experiments/exp-085_generational_transmission/` and `writing/preprints/2026-07-21_generational_transmission/`. Prior-rung data: `exp-062_corpus_statistics/`, `exp-084_pcfg_discriminator/`.

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